Task A

Initial Schema and Functional Dependencies:

PRODUCT

PRODUCT(ProductID, ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID)

- **Primary Key**: ProductID
- **Foreign Key**: CategoryID → CATEGORY.CategoryID
- **Functional Dependency**:
    - ProductID → ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID
- **3NF Status**: In 3NF because the determinant, ProductID, is a candidate key.

CATEGORY

CATEGORY(CategoryID, CategoryName, Description)

- **Primary Key**: CategoryID
- **Candidate Keys**: CategoryID, CategoryName (unique)
- **Foreign Key**: None
- **Functional Dependencies**:
    - CategoryID → CategoryName, Description
    - CategoryName → CategoryID, Description
- **3NF Status**: In 3NF because each determinant is a candidate key.

INVENTORY

INVENTORY(InventoryID, ProductID, Quantity, MinStock, LastUpdated)

- **Primary Key**: InventoryID
- **Candidate Keys**: InventoryID, ProductID (unique)
- **Foreign Key**: ProductID → PRODUCT.ProductID
- **Functional Dependencies**:
    - InventoryID → ProductID, Quantity, MinStock, LastUpdated
    - ProductID → InventoryID, Quantity, MinStock, LastUpdated
- **3NF Status**: In 3NF because each determinant is a candidate key.

SALE

SALE(SaleID, SaleDate, TotalCost, TotalRevenue)

- **Primary Key**: SaleID
- **Foreign Key**: None
- **Functional Dependency**:
    - SaleID → SaleDate, TotalCost, TotalRevenue
- **3NF Status**: In 3NF because SaleID is a candidate key.

SALEITEM

SALEITEM(SaleItemID, SaleID, ProductID, QuantitySold, UnitPrice, UnitCost)

- **Primary Key**: SaleItemID
- **Foreign Keys**:
    - SaleID → SALE.SaleID
    - ProductID → PRODUCT.ProductID
- **Functional Dependency**:
    - SaleItemID → SaleID, ProductID, QuantitySold, UnitPrice, UnitCost
- **3NF Status**: In 3NF because SaleItemID is a candidate key.

Final 3NF Schema

All initial relations satisfy 3NF, so no decomposition was required. The final schema is unchanged.

PRODUCT(ProductID, ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID)

- PK: ProductID
- FK: CategoryID → CATEGORY.CategoryID

CATEGORY(CategoryID, CategoryName, Description)

- PK: CategoryID
- Unique: CategoryName

INVENTORY(InventoryID, ProductID, Quantity, MinStock, LastUpdated)

- PK: InventoryID
- FK: ProductID → PRODUCT.ProductID
- Unique: ProductID

SALE(SaleID, SaleDate, TotalCost, TotalRevenue)

- PK: SaleID

SALEITEM(SaleItemID, SaleID, ProductID, QuantitySold, UnitPrice, UnitCost)

- PK: SaleItemID
- FK: SaleID → SALE.SaleID, ProductID → PRODUCT.ProductID